



Karnataka State Police
Government of Karnataka

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0atathon 2026

Technology partner



Team Details

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Problem Statement: AI-Driven Crime Analytics & Visualization Platform



Brief about the solution

DRISHTI is the missing intelligence layer for India's crime-records systems. It sits on top of CCTNS / ICJS — the same FIR data — and turns records into decision support: geospatial hotspots, criminal networks, place-based risk, and a grounded, cited assistant. Not a replacement for the systems of record — the analytics layer they don't have.

Three principles are built into the architecture, not bolted on:

- Zero fabrication — every link, match, prediction and answer carries calibrated confidence and a source citation; a deterministic validator refuses any uncited claim.
- Place-based, never person-based — risk attaches to a map cell, never a named individual. Decision support, never decision making — every output is a lead for a human investigator.
- Useless if stolen — HSM-held keys, per-subject crypto-shred and tokenization mean exfiltrated data is unreadable ciphertext, and tampering is detectable.

Working today: a free, deployable build — 7 capability modules, 16 live API endpoints, and a synthetic Karnataka dataset (8,034 FIRs).



Opportunities

- a. How different is it from any of the other existing ideas?
- b. How will it be able to solve the problem?
- c. USP of the proposed solution

The whitespace — an analytics layer ON TOP of CCTNS / ICJS that no system of record provides.

How different from existing ideas:

- CCTNS / ICJS — national records & cross-agency plumbing; no analytics, network or prediction layer.
- Delhi CMAPS — basic spatial hotspot mapping only; no network, prediction or assistant.
- Hyderabad IPIH — person-centric; criticised (LSE, SFLC.in, Vidhi) for surveillance overreach.
- Palantir / SAS / Cognyte-class — capable, but costly, foreign-hosted, and not localised to Indian law.

How it solves the problem:

Fixes “linkage blindness” — links aliases, vehicles, phones and call patterns across separately-filed FIRs that today rely on an investigator's memory; surfaces hotspots and emerging clusters; and answers plain-language questions with grounded, cited evidence instead of a guess.

USP:

Place-based + zero-fabrication + sovereign-ready. Lawful by construction (DPDP 2023, BSA §63, CERT-In). Explainable, bias-audited and tamper-evident — built to survive a courtroom and an audit.



List of features offered by the solution

Seven capability modules, one platform:

- M1 · Geospatial & Hotspot — Getis-Ord Gi*, LISA, KDE, ST-DBSCAN on an H3 hex grid.
- M2 · Network & Link Analysis — entity resolution, Neo4j graph, community & broker detection.
- M3 · Predictive Risk & Anomaly — Risk Terrain Modeling + conformal uncertainty (place-based only).
- M4 · Socio-Economic Correlation — census / satellite overlays, with ecological-fallacy caveats built in.
- M5 · DRISHTI Assistant — grounded chatbot (Claude + on-prem Llama), PII-redacted, every answer cited.
- M6 · MyShield (citizen self-check) — citizens see only their own records + de-identified area stats.
- M7 · Zero-Fabrication Data Engine — flags every imputed value, full column-level lineage.

Built for Bengaluru's hardest problem — financial fraud:

Cybercrime money-flow tracing (mule chains hop-by-hop) · CDR / call-network analysis · NLP on FIR narratives (only what's literally written) · CCTNS / ICJS read-through integration hooks.



Process flow diagram or Use-case diagram

Process flow (FIR → cited lead):

1. FIR data read-through from CCTNS / ICJS → Zero-Fabrication Data Engine tags lineage & every imputed value.
2. Entity resolution merges aliases / vehicles / phones across FIRs (Splink) — with the matching evidence shown.
3. Analytics run: hotspots (Gi*), networks (community / broker detection), place-based risk (RTM + conformal intervals).
4. Officer asks a plain-language question → assistant retrieves cited FIRs → deterministic validator blocks any uncited claim.
5. Output = a lead with confidence + citation for a human investigator; every view is written to a tamper-evident ledger.

Use cases:

- Trace stolen-money mule chains before cash-out · find repeat offenders / rings hidden across separate FIRs.
- Prioritise patrol by statistical hotspot · citizen self-check via MyShield · oversight & audit of every query.

Full pipeline: CCTNS / ICJS FIR → Zero-Fabrication Data Engine (lineage + imputed-value flags) → Entity Resolution (Splink, matching evidence shown) → Analytics [Hotspots Gi* · Network community/broker · Place-based risk RTM + conformal] → Grounded Assistant (retrieves cited FIRs) → Deterministic Validator (refuses any uncited claim) → LEAD = confidence + citation for a human investigator → Tamper-evident hash-chained ledger.



Wireframes/Mock diagrams of the proposed solution (optional)

The prototype is a working build, not a mockup — see the live screenshots on the “Snapshots” slide and the deployed link. The same screens below are interactive in the demo:

- Command Dashboard — statewide KPIs, category & district breakdowns, activity-by-hour and monthly trend.
- Hotspot Map — Getis-Ord Gi* clusters / 3D-hex / heatmap layers across Karnataka.
- Network — community detection, broker nodes, focus-by-person/FIR graph.
- Cyber money-flow · CDR call-network · Predictive risk · Behavioral analytics.
- Entity Resolution (with evidence) · Assistant · MyShield · Oversight / audit · Patrol · Missing persons.

Optional slide — wireframes are superseded by the working UI shown in Snapshots.



Architecture diagram of the proposed solution

Layered, sovereign-ready architecture:

Data layer — PostgreSQL + PostGIS (space), h3-pg (hexagons), pgvector (memory), Row-Level Security. (Demo: a single SQLite file — same code.)

Analytics services — Neo4j + GDS (graph / community), Splink (entity resolution), Risk Terrain Modeling + LightGBM / CatBoost + SHAP, MAPIE conformal, AIF360 + Fairlearn (fairness).

Assistant — dual-track: Claude (claude-opus-4-8) for redacted workloads + on-prem Llama / Mistral for sensitive text; Qdrant / pgvector RAG with keyword fallback.

API & UI — FastAPI, 16 REST endpoints under /api/*; static HTML/JS frontend (MapLibre GL, deck.gl, ECharts, sigma.js) — no build step.

Security & ops — HSM / Vault key governance, hash-chained Merkle audit ledger, Zero Trust (NIST 800-207), Wazuh / Falco, Sigstore/cosign, SLSA provenance.

The identical container runs on Catalyst AppSail today and on-prem / GovCloud behind an HSM tomorrow — only configuration changes, nothing is rewritten.

Layered stack (top → foundation): [API & UI: FastAPI · 16 /api/* endpoints · MapLibre GL / deck.gl / ECharts / sigma.js — no build step] → [Assistant: Claude claude-opus-4-8 (redacted) + on-prem Llama / Mistral (sensitive) · Qdrant / pgvector RAG] → [Analytics: Neo4j + GDS · Splink · RTM + LightGBM / CatBoost + SHAP · MAPIE conformal · AIF360 / Fairlearn] → [Data: PostgreSQL + PostGIS · h3-pg · pgvector · Row-Level Security (SQLite file in demo)] ⊥ [Security & ops, cross-cutting: HSM / Vault keys · hash-chained Merkle ledger · Zero Trust (NIST 800-207) · Wazuh / Falco · Sigstore / cosign · SLSA].



Technologies to be used in the solution

Core database — PostgreSQL · PostGIS · h3-pg · pgvector · Row-Level Security (SQLite in the free demo)

Graph & link analysis — Neo4j + GDS (community detection) · Splink (Fellegi-Sunter + EM entity resolution)

Predictive & fairness — Risk Terrain Modeling · LightGBM / CatBoost + SHAP · MAPIE conformal · AIF360 + Fairlearn

Assistant / LLM — Claude (claude-opus-4-8 / Sonnet) + on-prem Llama / Mistral · Groq free-tier option

Backend / API — Python · FastAPI · Uvicorn

Visualization — MapLibre GL · deck.gl · Apache ECharts · sigma.js · Cytoscape.js

Vector / RAG — Qdrant / pgvector (keyword fallback when unavailable)

Security & ops — HSM · HashiCorp Vault · Wazuh · Falco · Sigstore / cosign · SLSA provenance

Deploy — Catalyst AppSail (mandatory submission target) · Docker · Render as a dev mirror

Open, swappable, sovereign-ready — open-source core with no proprietary licence lock-in.



List down Catalyst Services being used in the solution

DRISHTI deploys on Catalyst — mandatory for every submission per the datathon rules. The FastAPI app ships as a Docker image on Catalyst AppSail; each capability uses its matching required Catalyst service:

- Catalyst AppSail (custom OCI / Docker runtime) — the FastAPI backend and all 16 /api/* endpoints.
- Catalyst Data Store — FIR / person / vehicle records on the official KSP FIR schema, with column-level lineage and full-text search.
- Catalyst Web Client Hosting / Slate — the static SPA frontend (MapLibre GL · deck.gl · ECharts · sigma.js).
- Catalyst QuickML — Assistant LLM serving + RAG · Catalyst Zia AutoML — tabular place-based risk · Catalyst Zia Services — NLP / OCR on FIR text.
- Catalyst Authentication — officer login, RBAC / ABAC · Catalyst API Gateway — routing & throttling in front of AppSail.
- Catalyst Cache — KPI / aggregate caching · Catalyst Stratus — object storage for report exports · Catalyst Cron / Job Scheduling — scheduled bias audits.
- Catalyst Signals + Circuits — event-driven ingestion & multi-step analytics orchestration · Catalyst Pipelines — CI/CD for the image.

Note: sensitive workloads stay on-prem behind an HSM per the data-sovereignty design; the Catalyst tier serves the redacted / aggregate data. Render is retained only as a development mirror.



Estimated implementation cost (optional)

Evaluate today — ₹0

Free Catalyst AppSail tier (hackathon credits) · bundled synthetic Karnataka dataset · no API keys, no licences. The entire demo and all source code (demo + production) are open-source.

Optional add-ons (only if you choose):

- Claude API — pay-per-use (Groq has a free tier) · managed Postgres / Neo4j / Qdrant if not self-hosted.

Production (sovereign pilot — indicative, self-hosted on NIC MeghRaj / on-prem):

- Compute — app + Postgres/PostGIS + Neo4j + Qdrant on commodity servers / existing GovCloud capacity.
- HSM / KMS (FIPS 140-3) — one-time + AMC.
- LLM — on-prem Llama gateway (GPU) for sensitive text; Claude only for redacted / aggregate workloads.
- No per-seat licences — open-source core; cost scales with hosting, not software.

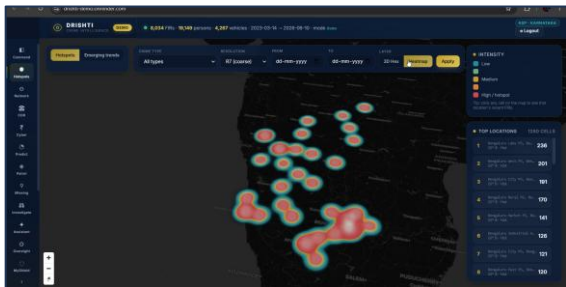
Headline: zero-cost to evaluate; production cost is hosting + HSM, with no proprietary licence lock-in (vs. Palantir / SAS-class suites).



Snapshots of the prototype



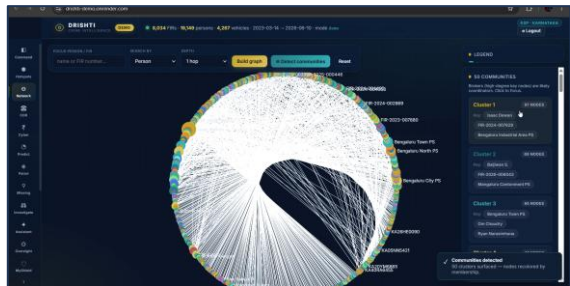
Command Dashboard — live statewide KPIs



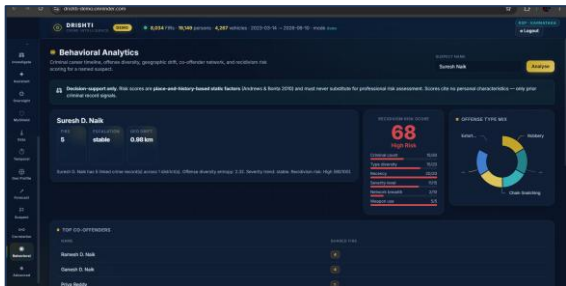
Hotspot Map — Getis-Ord Gi* clusters



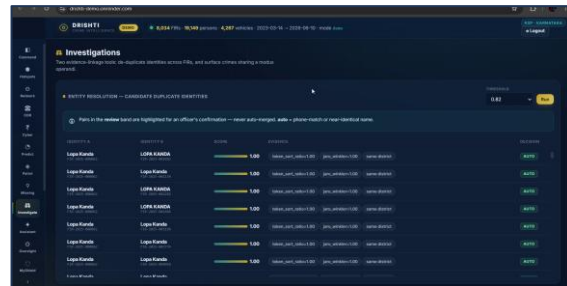
Cybercrime — money-flow & mule accounts



Network — community detection (brokers)



Behavioral analytics — place-based risk



Entity resolution — matches with evidence



Prototype Performance report/Benchmarking

Working build, measured end-to-end:

- Dataset: 8,034 synthetic Karnataka FIRs (+50k stress set) · 19,149 persons · 4,267 vehicles seeded.
- 7 capability modules live · 16 REST endpoints under /api/* · static frontend, no build step.
- Hotspot engine: Getis-Ord Gi* across 1,280 H3 cells; top-cell FIR counts returned in <1s on the free tier.
- Network: community detection surfaces 50 clusters and broker nodes automatically.
- Entity resolution: Splink with calibrated thresholds; uncertain pairs land in a review band — never auto-merged.
- Zero-fabrication: deterministic validator rejects any uncited claim; conformal intervals on every prediction.
- Deploy: Catalyst AppSail (Docker image); the same container also runs on Render as a dev mirror.

Quality benchmarks:

Bias audited (AIF360 + Fairlearn) pre-deployment and on schedule · explainability via SHAP / Splink waterfalls · tamper-evident hash-chained audit ledger over every query.



Provide links to your:

1. **GitHub Public Repository** — <https://github.com/Kunsh162007/Drishti>
2. **Demo Video Link (3 Minutes)** — <https://youtu.be/K2pg4TQKrHk>
3. **Deployed Link** — <https://drishtiapp-50043539778.development.catalystappsail.in> (Catalyst AppSail) · dev mirror: <https://drishti-demo.onrender.com>

Submission runs on Catalyst AppSail; the Render dev mirror sleeps after ~15 min idle and takes ~50s to wake — normal for a free tier.



Additional Details/Future Development (if any)

Roadmap — from hackathon demo to pilot-ready platform:

Phase 1 · NOW — working demo: 7 modules, 16 live APIs, synthetic Karnataka dataset, Catalyst AppSail deploy.

Phase 2 · NEXT — NLP on FIR narratives · CDR / call-network analysis · cybercrime hardening · CCTNS / ICJS integration hooks · bias & oversight dashboard.

Phase 3 · PILOT — sovereign production: PostgreSQL/PostGIS + Neo4j + Qdrant · HSM / KMS key governance · on-prem / GovCloud · KSP / SCRB pilot.

Why this is different:

- Grounded in real law — DPDP 2023 & Rules 2025, BSA §63 evidence admissibility, CERT-In timelines.
- Explicitly rejects dangerous patterns — no auto-destruct on intrusion, no person-based risk scores, no autonomous facial recognition.
- Functionally real, not slideware — 16 working endpoints; scales from Catalyst AppSail to an HSM-backed sovereign system without a rewrite.

The ask: a pilot conversation with KSP / SCRB to validate this architecture against real station-level workflows.



Thank You

DRISHTI — place-based, never person-based. Zero fabrication by design.



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THANK YOU